

Convex Optimization 作业 4

December 2025

1 Given function $f(x) = -5x^2 + 5x$, work out its conjugate function and plot it out.

(1) 在凸分析里, 共轭函数 (也叫 Fenchel 共轭) 定义为: $f^*(y) = \sup_{x \in \mathbb{R}} (yx - f(x))$

此处 $f^*(y) = \sup_{x \in \mathbb{R}} (yx - 5x + 5x^2)$

(2) 对 $f^*(y)$ 求对 x 的偏导得 $f^*(y)' = 10x + (y - 5)$

(3) 驻点方程 $10x + (y - 5) = 0 \Rightarrow x = \frac{5-y}{10}$

(4) 代入 $f^*(y)$ $f^*(y) = -\frac{(5-y)^2}{20}$

(5) 图形如下

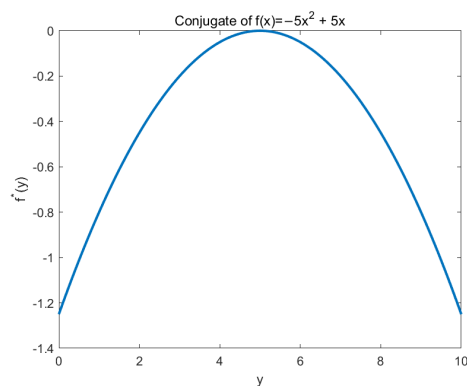


图 1: Conjugate of $f(x) = -5x^2 + 5x$

2 Suppose $f : S \subset \mathbb{R} \rightarrow \mathbb{R}$ is convex. Let $a, b \in S$ and $a < b$. Show that

$$f(c) \leq \frac{b-c}{b-a}f(a) + \frac{c-a}{b-a}f(b), \quad \forall c \in [a, b].$$

(1) 凸函数定义, 对任意 $x_1, x_2 \in S$ 和任意 $\lambda \in [0, 1]$,

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2).$$

(2) 取 $a, b \in S$, 且 $a < b, c \in [a, b]$, 令

$$\lambda = \frac{b-c}{b-a}, \quad 1-\lambda = \frac{c-a}{b-a}.$$

且 $c = \lambda a + (1-\lambda)b$

(3)

$$f(c) = f(\lambda a + (1-\lambda)b) \leq \lambda f(a) + (1-\lambda)f(b) = \frac{b-c}{b-a}f(a) + \frac{c-a}{b-a}f(b).$$

3 Given functions $f(x)$ and $g(x)$ are convex in $\text{dom } f$ and $\text{dom } g$, prove that $h(x) = f(x) + g(x)$ is convex in $\text{dom } h$.

(1) 因为 $f(x)$ 和 $g(x)$

$$f(\lambda \alpha_1 + (1-\lambda)\alpha_2) \leq \lambda f(\alpha_1) + (1-\lambda)f(\alpha_2)$$

$$g(\lambda \alpha_1 + (1-\lambda)\alpha_2) \leq \lambda g(\alpha_1) + (1-\lambda)g(\alpha_2).$$

(2) 对和函数 $h = f + g$ 写出同一凸组合

$$\begin{aligned} h(\lambda \alpha_1 + (1-\lambda)\alpha_2) &= f(\lambda \alpha_1 + (1-\lambda)\alpha_2) + g(\lambda \alpha_1 + (1-\lambda)\alpha_2) \\ &\leq \lambda f(\alpha_1) + (1-\lambda)f(\alpha_2) + \lambda g(\alpha_1) + (1-\lambda)g(\alpha_2) \\ &= \lambda(f(\alpha_1) + g(\alpha_1)) + (1-\lambda)(f(\alpha_2) + g(\alpha_2)) \\ &= \lambda h(\alpha_1) + (1-\lambda)h(\alpha_2). \end{aligned}$$

4 Prove that the function

$$f(x, y) = x \log(x) + y \log(y), \quad x > 0, y > 0$$

is convex

(1) 计算一阶偏导

$$\frac{\partial f}{\partial x} = 1 + \log x, \quad \frac{\partial f}{\partial y} = 1 + \log y.$$

(2) 计算二阶偏导 (Hessian 元素)

$$\frac{\partial^2 f}{\partial x^2} = \frac{1}{x}, \quad \frac{\partial^2 f}{\partial y^2} = \frac{1}{y}, \quad \frac{\partial^2 f}{\partial x \partial y} = 0.$$

(3) Hessian 矩阵

$$\nabla^2 f(x, y) = \begin{bmatrix} \frac{1}{x} & 0 \\ 0 & \frac{1}{y} \end{bmatrix}.$$

(4) 正定性对任意 $x > 0, y > 0$ 有

$$\frac{1}{x} > 0, \quad \frac{1}{y} > 0, \quad \det(\nabla^2 f) = \frac{1}{xy} > 0,$$

故 Hessian 正定，函数严格凸。

$$f(x, y) = x \log x + y \log y \quad \text{在 } x > 0, y > 0 \text{ 上严格凸。}$$

5 Convert the following Lasso regression model into a standard QP problem without changing the objective function.

$$\text{Min.} \quad \|y - \theta^\top X\|_2^2$$

$$\text{s.t.} \quad \|\theta\|_1 \leq t$$

$$y \in \mathbb{R}, \quad X \in \mathbb{R}^{n+1}, \quad \theta \in \mathbb{R}^{n+1}$$

where t is a positive constant.

将目标函数展开得到

$$\|y - X\theta\|_2^2 = (y - X\theta)^T(y - X\theta) = y^T y - 2y^T X\theta + \theta^T X^T X\theta$$

$y^T y$ 是常数，在最小化时可以忽略

令 $\theta = \theta^+ - \theta^-$, 其中, $\theta^+ \geq 0, \theta^- \geq 0$:

$$\begin{aligned} \min_{\theta^+, \theta^-} \quad & \frac{1}{2} \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix}^T Q \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} + c^T \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} \\ \text{s.t.} \quad & A \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} \leq b \end{aligned}$$

其中

$$Q = \begin{bmatrix} XX^T & -XX^T \\ -XX^T & XX^T \end{bmatrix},$$

$$c = \begin{bmatrix} -2yX^T \\ 2yX^T \end{bmatrix},$$

$$A = \begin{bmatrix} 1, 1 \end{bmatrix},$$

$$b = t$$

最后答案

$$\begin{aligned} \min_{\theta^+, \theta^-} \quad & \frac{1}{2} \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix}^T \begin{bmatrix} XX^T & -XX^T \\ -XX^T & XX^T \end{bmatrix} \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} + \begin{bmatrix} -2yX^T \\ 2yX^T \end{bmatrix}^T \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} 1, 1 \end{bmatrix} \begin{bmatrix} \theta^+ \\ \theta^- \end{bmatrix} \leq t \end{aligned}$$